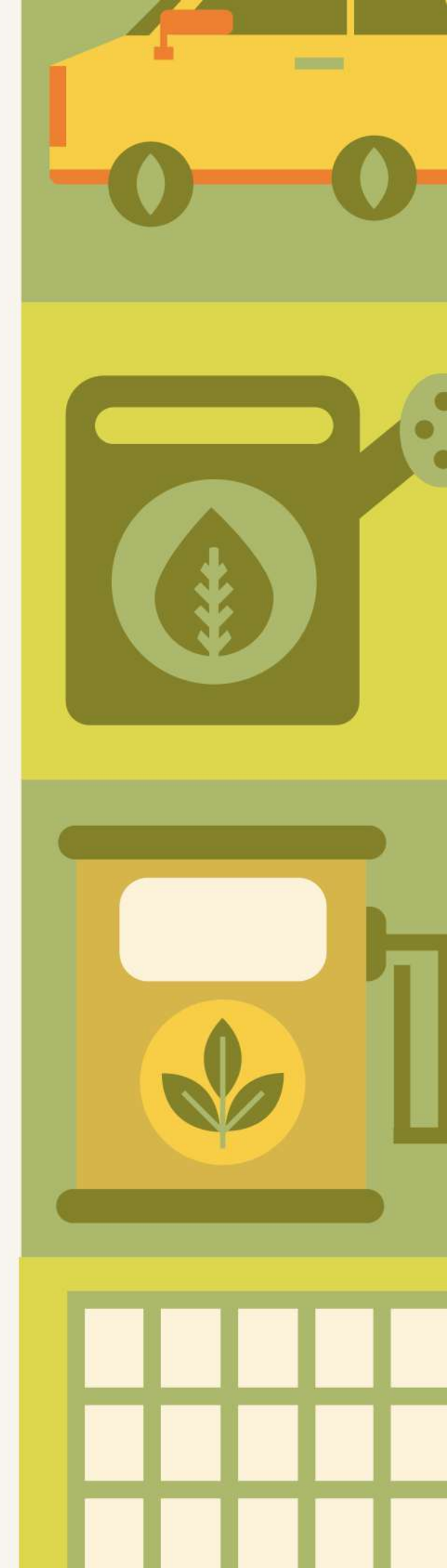




DEVELOPMENT OF GEOPOLYMER ZEOLITES FOR EFFICIENT AND SUSTAINABLE CARBON CAPTURE

Theme: Sustainable Energy

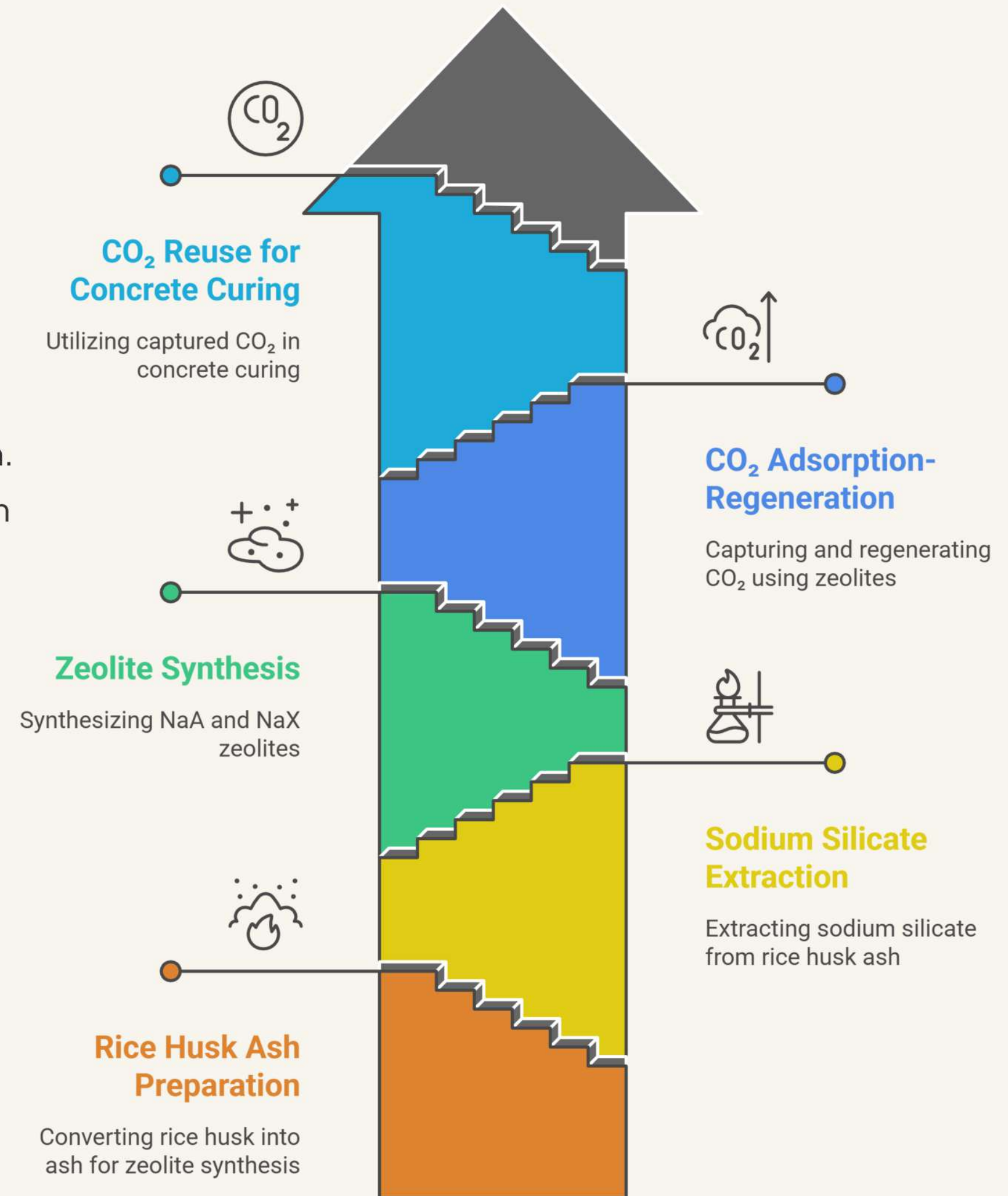


Executive Summary

Our project proposes an innovative, sustainable solution for industrial carbon capture using geopolymer zeolites synthesized from rice husk, an abundant agricultural waste in Northeast India. Through a four-stage process—rice husk ash preparation, sodium silicate extraction, zeolite synthesis, and CO₂ adsorption-regeneration—we convert biomass waste into NaA and NaX zeolite adsorbents with high CO₂ affinity.

Captured CO₂ is reused for concrete curing, forming a closed carbon utilization loop that minimizes emissions while adding value to waste materials.

This approach integrates materials science and environmental engineering to provide a low-cost, scalable, and energy-efficient carbon capture alternative that can reduce capture costs by up to 50%, utilize local agricultural residues, and significantly contribute to India's Net Zero 2070 vision.



Problem Statement

Context

Industrial CO₂ emissions are one of the leading contributors to global warming, accounting for over 35% of total greenhouse gases.

At the same time, India generates 25–30 million tons of rice husk annually, much of which is openly burned, releasing CO₂ and particulate matter into the atmosphere. Northeast India, a major rice-producing region, faces both problems biomass waste mismanagement and industrial carbon emissions without a viable, affordable carbon capture solution.

Pain Point

Industries need an affordable, sustainable, and regenerable adsorbent that captures CO₂ effectively under mild conditions while supporting circular waste reuse.

Research Insights

- Studies show fly ash and rice husk ash can yield high-silica aluminosilicate materials suitable for zeolite formation.
- Geopolymer zeolites combine the strength of geopolymers with the adsorption performance of zeolites, reducing both CO₂ capture costs and waste disposal impacts.

Market Opportunity

The global carbon capture, utilization, and storage (CCUS) market is projected to exceed USD 15 billion by 2030, with India among the top emerging adopters.

Developing rice husk-based geopolymer zeolites offers a regionally relevant, low-cost, and scalable solution to enter this growing market.

Who is affected

- Industries: Power, cement, and steel sectors struggling with costly carbon capture methods.
- Communities: Facing air pollution from biomass burning and climate impacts from industrial emissions.
- Government & Policy Stakeholders: Seeking scalable solutions to meet India's Net Zero 2070 commitments.

Proposed Idea

Our solution converts rice husk - an abundant agricultural waste in Northeast India—into high-performance zeolite adsorbents for industrial CO₂ capture. The process works through four stages

Rice Husk Ash Preparation

- Collect rice husk from local mills.
- Wash the rice husk.
- Dry it at 110°C
- Calcinate at 450°C for 6 hours to produce rice husk ash (RHA).

Sodium Silicate Extraction

- Treat RHA with acidic solution to remove impurities.
- React with sodium hydroxide solution at 90°C for 5 hours to extract sodium silicate.

Zeolite synthesis

- Grind dry sodium silicate & sodium aluminate.
- By hydrothermal method, heat at 100°C for 8 hours.
- This produces NaA & NaX

Carbon Capture & Regeneration

- CO₂ is captured using adsorbent beds of zeolites.
- Regenerate zeolite using temperature/pressure swing.
- Use the released CO₂ for concrete curing.

Innovation and Creativity

What Makes it Innovative?

1. Waste-to-Resource Pathway:

Uses rice husk, a regional agricultural waste, as the primary silica source — turning pollution into value.

2. Material Innovation:

Creates hybrid geopolymer-zeolite materials combining strength, reusability, and high CO₂ affinity.

3. Low-Energy Synthesis:

Operates under mild hydrothermal conditions (~100°C) with no high-temperature calcination, reducing energy cost by >50%.

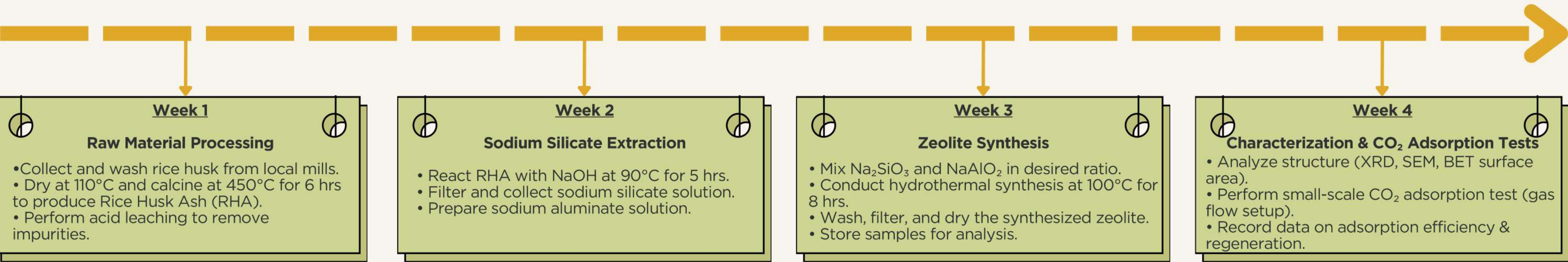
4. Closed Carbon Cycle:

Reuses captured CO₂ for concrete curing, creating a circular carbon loop between agriculture, industry, and construction.

What Makes it Unique?

<u>Parameter</u>	<u>Traditional Zeolite</u>	<u>Rice Husk Geopolymer Zeolite</u>
Raw Material	Synthetic silica	Rice Husk Ash (waste)
Synthesis Temp	200–300°C	~100°C
Cost	High	Low
Sustainability	Moderate	Excellent
CO ₂ Adsorption	High	Comparable or better
Regeneration Energy	High	Low

Implementation Plan



Resources Required

Category	Details
Materials	Rice husk, NaOH, NaAlO ₂ , acids (HCl/H ₂ SO ₄), distilled water
Equipment	Furnace, drying oven, pH meter, hydrothermal reactor, CO ₂ gas setup
Lab Facilities	Materials or environmental engineering lab with adsorption setup
Team Expertise	Materials synthesis, chemical handling, adsorption testing

Potential Risks and Mitigation

Risk	Mitigation
Incomplete silica extraction	Optimize NaOH concentration and reaction time
Non-crystalline zeolite formation	Maintain precise temperature/time during hydrothermal synthesis
Low CO ₂ adsorption	Adjust Si/Al ratio and activation conditions
Equipment limitation	Use available lab-scale hydrothermal setup or autoclave

Rethinking CO₂ Capture

Method	Cost	Energy Demand	Moisture Sensitivity	Scalability
MEA Scrubbing	High	Very High	Moderate	Medium
Synthetic Zeolites	Medium	Moderate	High	High
Cryogenic Separation	Very High	Very High	Very High	Low
Membrane Separation	High	Moderate	High	Medium
Biochar Sequestration	Low	Moderate	Low	Low
Geopolymer Zeolites	Low	Low	Low-Moderate (tunable)	High (modular + local)

WHY OUR GEOPOLYMER ZEOLITES STAND OUT

-  MADE FROM AGRI-WASTE (RICE HUSK) — LOW COST, SUSTAINABLE
-  LOWER ENERGY USE (~100°C SYNTHESIS) THAN MEA OR CRYOGENICS
-  DESIGNED FOR MOISTURE RESISTANCE AND MODULAR SCALE-UP

Impact of the Geopolymer Zeolite Carbon Capture Solution

Environmental Impact

- Captures up to 1,000 tons CO₂/year at pilot scale
- CO₂ reused in concrete curing (up to 80% retained)
- Prevents agricultural waste burning (rice husk → adsorbent)
- Cuts carbon capture energy by ~50% vs MEA scrubbing

Economic & Social Impact

- 10+ jobs created in rural operations
- Adds value to 25–30 tons/month of agri-waste
- Estimated ₹72+ lakhs/year revenue (zeolite sales + carbon credits)
- Boosts rural income via husk procurement and processing

Tracking and Validation

- Flue gas analyzers for real-time CO₂ monitoring
- XRD and SEM for post-capture verification
- Concrete tests to measure strength, durability, and retained CO₂
- Enables data-backed scaling & reporting

FINANCIAL ANALYSIS

CAPITAL EXPENDITURE (CAPEX)

Detailed Equipment & Infrastructure Costs

Item	Cost (₹)	% of Total
Calcination Furnace (450°C)	8,50,000	18.6%
CO ₂ Adsorption System	7,00,000	15.3%
Hydrothermal Reactor (100°C)	6,00,000	13.1%
Facility Setup & Civil Work	5,50,000	12.0%
Grinding & Milling Equipment	4,50,000	9.8%
Drying Equipment (110°C)	3,50,000	7.6%
Storage Tanks & Materials	2,50,000	5.5%
Safety Systems & Controls	2,50,000	5.5%
Installation & Commissioning	2,00,000	4.4%
Contingency Fund (10%)	3,80,000	8.3%
TOTAL CAPEX	₹45,80,000	100%

Additional Initial Costs

- Working Capital (2 months): ₹11.40 Lakhs
- Regulatory & Permits: ₹1.00 Lakhs
- Pre-operational Expenses: ₹1.50 Lakhs
- TOTAL INITIAL INVESTMENT: ₹59.70 Lakhs

OPERATING EXPENDITURE (OPEX)

(Monthly Operating Costs)

A. Raw Materials (₹72,500/month)

Item	Quantity	Unit Cost	Monthly Cost
Rice Husk	5 tons	₹2,000/ton	₹10,000
Sodium Hydroxide (NaOH)	400 kg	₹45/kg	₹18,000
Sodium Silicate	350 kg	₹40/kg	₹14,000
Sodium Aluminate	130 kg	₹50/kg	₹6,500

B. Energy & Utilities (₹85,000/month)

Item	Consumption	Rate	Monthly Cost
Electricity	7,500 kWh	₹6.5/kWh	₹48,750
Water	-	-	₹3,250

C. Labor Costs (₹565,000/month)

Position	Number	Salary/Month	Total Cost
Plant Operators	8 (2 shifts)	₹28,000	₹2,24,000
Supervisors	2	₹45,000	₹90,000
Maintenance Technician	1	₹32,000	₹32,000
QC Lab Technician	1	₹32,000	₹32,000

D. Fixed Operating Costs (₹75,000/month)

- Maintenance & Repairs: ₹28,000
- Quality Testing & Analysis: ₹15,000
- Insurance: ₹8,500

FINANCIAL ANALYSIS

Annual Operating Summary

Category	Monthly	Annual
Raw Materials	₹48,500	₹5,82,000
Energy & Utilities	₹52,000	₹6,24,000
Labor	₹3,58,000	₹42,96,000
Fixed Costs	₹51,500	₹6,18,000
Marketing & Admin	₹20,000	₹2,40,000
TOTAL OPEX	₹5,30,000	₹63,60,000

Note: Government subsidy of ₹4.80L in Year 1 reduces effective OPEX to ₹58.80L

Pricing Strategy & Market Assumptions

- Carbon Credits: ₹3,200/ton CO₂ (voluntary); 10% annual growth; VCS certified.
- Zeolite Adsorbent: ₹125/kg (premium), ₹110/kg (>1 ton); target: thermal power & cement industries.
- CO₂ Utilization: ₹750/ton (concrete curing); 50% CO₂ sold to construction firms.

Revenue Breakdown (Year 1 - 50% Capacity)

Revenue Source	Units/Year	Price	Annual Revenue	% Share
Carbon Credits	1,000 tons CO ₂	₹3,200/ton	₹32,00,000	44%
Zeolite Adsorbent Sales	25,000 kg	₹125/kg	₹31,25,000	43%
CO ₂ for Concrete Curing	500 tons	₹750/ton	₹3,75,000	5%
Government Grants/Subsidies	-	-	₹4,80,000	7%
Technical Consulting	-	-	₹20,000	1%
TOTAL REVENUE (Year 1)	-	-	₹72,00,000	100%

FIVE-YEAR FINANCIAL PROJECTIONS

Year	Capacity	Revenue	OPEX	Gross Profit	Net Profit*	Cumulative Profit
Year 1	50%	₹72,00,000	₹63,60,000	₹8,40,000	₹3,60,000	₹3,60,000
Year 2	70%	₹1,12,00,000	₹71,21,600	₹40,78,400	₹36,18,400	₹39,78,400
Year 3	85%	₹1,48,00,000	₹79,49,296	₹68,50,704	₹63,90,704	₹1,03,69,104
Year 4	95%	₹1,75,00,000	₹86,59,451	₹88,40,549	₹83,20,549	₹1,86,89,653
Year 5	100%	₹2,10,00,000	₹93,67,801	₹1,16,32,199	₹1,10,52,199	₹2,97,41,852

FINANCIAL ANALYSIS

KEY FINANCIAL ASSUMPTIONS

Production Assumptions

Parameter	Assumption	Basis
Rice husk cost	₹2,000/ton	Partnership with local mills (subsidized)
Rice husk availability	150 tons/month	Network of 8 mills within 40km
Conversion rate	1 ton husk → 220 kg RHA	Optimized process
Zeolite yield	1 kg RHA → 2 kg zeolite	Enhanced synthesis method
Production days	310 days/year	High reliability operations
Plant uptime	90%	Modern equipment, preventive maintenance

Market Assumptions

Parameter	Year 1	Year 3	Year 5
Carbon credit price	₹3,200/ton	₹3,600/ton	₹4,200/ton
Zeolite price	₹125/kg	₹120/kg	₹115/kg
Market growth rate	25%/year	18%/year	12%/year
Competition impact	Minimal	Moderate	High
Customer retention	85%	90%	92%

Cost Optimization Assumptions

Cost Category	Year 1	Year 2	Year 3-5
Raw materials efficiency	Base	+8%	+12%
Energy efficiency	Base	+10%	+15%
Labor productivity	Base	+15%	+20%
Overall cost reduction	-	12%	18%

Performance Assumptions

Parameter	Assumption	Basis
CO ₂ capture capacity	3.5 mmol/g zeolite	Laboratory verified results
Adsorption efficiency	88% from flue gas	Pilot testing data
Regeneration cycles	85 cycles/batch	Before 15% degradation
Annual replacement	12% of adsorbent	Quality maintenance program

Government Scheme Assumptions

- Capital subsidy: 30% (₹13.74L already factored in reduced CAPEX)
 - Production-linked incentive: ₹4.80L
 - Tax holiday: 3 years (not shown in profit calculations)
 - Green energy certification benefits
- Institutional mechanism planned: a “Carbon Capture Finance Corporation of India” to support capital + operational costs of CCUS projects.
 - Incentives ranging from 50 % to 100 % funding support for certain CCUS pilot/industrial projects.

Scalability

1. Resource Availability

- India produces over 20 million tons of rice husk annually, ensuring an abundant and renewable feedstock supply.
- Local sourcing minimizes logistics costs and supports rural economies.

2. Modular Production Design

- The plant can be scaled up through modular reactor units, allowing gradual capacity expansion based on demand.
- Each module can function independently, enabling decentralized CO₂ capture near emission sources.

3. Cost-Effective Expansion

- Low-cost raw materials and simple synthesis steps (calcination, hydrothermal treatment) make replication easy across regions.
- Utilization of existing industrial waste infrastructure (fly ash handling, biomass plants) reduces capital barriers.

4. Market Readiness

- Can integrate with industrial emitters such as cement, steel, and power plants.
- High reusability of geopolymer zeolites ensures long-term operational sustainability and low lifecycle cost.



Timeline

week 1 & 2

Ideation & Planning

- Brainstorm concept & goals
- Allocate roles among team
- Research requirements & feasibility

- Prepare content/ materials
- Create posters, publicity plan
- Finalize logistics and resources

Design & Preparation

week 3 & 4

week 5

Execution / Main Event

- Conduct main event/activity
- Ensure smooth coordination
- Capture media coverage

- Collect participant feedback
- Analyze outcomes & performance
- Document learnings

Review & Feedback

week 6

week 7

Future Steps

- Plan continuation or improvement
- Submit final report or video summary

**THANK
YOU VERY
MUCH!**

